# Physics-Conditioned Video Diffusion with Kinematic Priors for Fusion Capsule Polishing

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## Abstract

Video diffusion models can generate realistic process videos, but using them as controllable surrogates for industrial systems remains difficult when key control variables are only partially observable and no simulator or differentiable physics model is available. We study this regime in inertial confinement fusion (ICF) target capsule polishing, where surface defects can degrade fusion yield. The dynamics are governed by the polishing speed $\omega_p$ and capsule-pad slip $S_C$. While capsule motion can be monitored from video, $S_C$ is only indirectly observable. Using only an analytical kinematic model, we adapt a frozen Stable Video Diffusion (SVD-xt) backbone via LoRA and condition the generation on Fourier-encoded physics parameters $(\omega_p, S_C)$. Two training objectives operate on latent frame-difference dynamics: (i) a batch level correlation loss aligning them with the analytical sliding speed, and (ii) a stratified ratio-matching loss that, conditioned on $\omega_p$, isolates slip structure from the dominant $\omega_p$-driven variance. We evaluate kinematic consistency by tracking capsule motion in generated videos, inverting the kinematic model, and measuring rank agreement under a physical validity gate. For speed, rank accuracy exceeds $0.97$ on real data; for slip, it reaches $0.735$ with Spearman $\geq 0.50$ on synthetic data. Overall, the results suggest a practical recipe for controllable industrial video surrogates when only low-dimensional analytical kinematics are available.

## 1 Introduction

Inertial confinement fusion (ICF) target capsules must be polished to nanometer tolerances because defects of only tens of nanometers can seed hydrodynamic instabilities and degrade ignition symmetry [23, 16, 1]. Polishing is therefore a costly throughput bottleneck, and improving it requires understanding how operating speed $\omega_p$ and capsule-pad slip $S_C$ drive material removal. These dynamics are only partly observable: the capsule moves stochastically inside an opaque fixture, and routine instrumentation consists of a top-down video stream.

We seek *controllable video surrogates*: given a real reference frame and prescribed $(\omega_p, S_C)$, the model samples plausible capsule-motion clips. Sampling supports counterfactual sweeps, uncertainty estimates for derived kinematics, and anomaly detection from the distribution of plausible outcomes. Analytical polishing kinematics [19, 20] also allows generated clips to be queried in physical coordinates: circulation, spin, and sliding speed. Most physics-guided video generation assumes PDE residuals, calibrated simulators, trajectory priors, dense motion supervision, or post-training guidance [3, 15, 39, 21, 43, 22, 34, 2, 37, 40]. Capsule polishing has none of these: the state is partly hidden, no calibrated video simulator exists for real footage, and the available physics is only a low-dimensional analytical map from $(\omega_p, S_C)$ to kinematic scalars. We therefore combine a stochastic video prior trained on real footage with scalar kinematic constraints.

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Our approach adapts a frozen Stable Video Diffusion (SVD-xt) [4] image-to-video backbone with LoRA. Fourier tokens encode $(\omega_p, S_C)$ and enter cross-attention. The kinematics define two weak supervision signals without absolute labels: a batch correlation loss $\mathcal{L}_{\text{dyn}}$ aligning latent frame differences with analytical sliding speed, and a stratified ratio loss $\mathcal{L}_{\text{ratio}}$ matching within-$\omega_p$ latent dynamics ratios to the analytical slip ratio. Because real $S_C$ is unobserved, we treat it as a prior-regularized latent conditioning variable and measure controllability by tracking generated clips and inverting them back to kinematics.

We combine these standard components to study controllable video generation under the sensing constraints of ICF capsule polishing. Our contributions are:

**(i) Video generation with analytical kinematic supervision.** We formulate a video surrogate that combines scalar conditioning with two latent objectives for analytical speed and slip relations. The formulation uses process videos and analytical kinematics without requiring a calibrated simulator of the real process or dense motion labels.

**(ii) Kinematic evaluation of generated motion.** We recover capsule circulation from generated pixels and use kinematic inversion to assess speed and slip ordering. We report rank accuracy, confidence intervals, and the fraction of clips passing validity checks alongside visual metrics. These scores measure consistency with the analytical model shared by training and evaluation.

**(iii) Speed control and the limits of slip recovery.** We distinguish observable speed control in real footage from slip ordering in synthetic clips with known labels. Across five evaluation seeds, the full model reaches real speed rank $0.985$ and synthetic slip rank $0.735$ (Tables 1 and 2). Real slip ordering remains at chance. Ablations show that scalar conditioning supplies most of the ordering, while the combined losses reduce synthetic FID-I3D and FVD without improving slip rank over conditioning alone.

## 2 Related Work

**Video diffusion and scalar conditioning.** Latent and video diffusion models provide strong short-horizon visual priors [26, 13, 4, 24], with LoRA enabling adaptation to small industrial datasets [14]. Continuous process controls require mechanisms that distinguish nearby physical values; common choices include FiLM, cross attention, ControlNet or IP-Adapter branches, and Fourier features [25, 41, 38, 29].

**Physics guided video generation.** Recent methods inject physics as field residuals or score guidance for known PDEs [3, 15], simulators for rigid bodies, materials, or interactions [39, 21, 5, 43, 34, 2], or dense optical-flow/trajectory signals [22, 35, 45]. Object-centric methods similarly assume spatial descriptors such as masks, contact fields, or skeleton tracks. Capsule polishing lacks these signals: the state is hidden, dense correspondences are unreliable on a featureless specular shell, and the only physics is the scalar pair $(\omega_p, S_C)$. We compare against flow and analytical warp baselines in physical coordinates (Sec. 6.5); dense-flow or trajectory-token baselines require supervision that is unavailable here.

**LoRA prediction and hypernetwork control.** Recent controllers can alter model weights or use scaling tokens: TC-LoRA predicts time-varying LoRA weights, Video2LoRA predicts LoRA modules from a reference video [6, 36], and LiON-LoRA-style fusion designs use tokens to modulate motion strength. Our controls are physical scalars supervised by kinematic ordering, so we keep one LoRA adapter set and inject Fourier operating tokens at every cross-attention layer. This gives a simple conditioning-only baseline analogous to an inference-time scaling-token controller: the "conditioning only" rows keep the tokens but remove $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$, isolating the marginal effect of the losses.

**Kinematic-consistency evaluation.** General video benchmarks for physical plausibility and scientific diffusion emulators emphasize that perceptual realism is not enough for physical use. Our metric follows that principle at the domain scale: instead of reporting only FID/FVD, we recover $\widehat{\omega_c}$ and $\widehat{R_T}$ from generated pixels and score monotonic agreement with operating variables. This does not make the videos mechanically exact, but it prevents appearance metrics from being mistaken for process controllability.

## 3 Preliminaries

**Problem.** Given a dataset $\mathcal{D} = \{(\mathbf{v}_i, \omega_p^{(i)}, S_C^{(i)}, \mathbf{x}_{\text{cond}}^{(i)})\}_{i=1}^N$ of clips $\mathbf{v}_i \in \mathbb{R}^{T\times 3\times H\times W}$, baseplate speed $\omega_p^{(i)}$, head-contact slip $S_C^{(i)}$, and first frame $\mathbf{x}_{\text{cond}}^{(i)}$, we learn $p_\theta(\mathbf{v} \mid \omega_p, S_C, \mathbf{x}_{\text{cond}})$ which clips are visually realistic and whose dynamics scale with $V_{\text{eff}}(\omega_p, S_C)$ (speed) and $R_T(S_C)$ (slip).

**Analytical kinematics.**

Following Lee's kinematic model [19, 20], a rigid ball of radius $R_b$ rests in a V-groove of half-angle $\theta$ and radius $R_g$ on a baseplate rotating at $\omega_p$. No slip at the two groove contacts and head-contact slip $S_C \in [0, 1]$ give the scalar map below (derivation in Appendix A1); Fig. 1 shows the contacts.
Two facts drive the losses: for fixed slip, $V_{\text{eff}}$ is linear in $\omega_p$ with zero intercept; for $\theta \in (0, \pi/2)$, $R_T$ is strictly increasing in $S_C$ and provides an ordinal slip target.

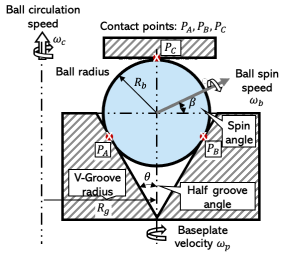



Figure 1: Polishing setup.

$$\omega_c = \omega_p \frac{1 + S_C \sin\theta}{1 + \sin\theta}, \quad \omega_b = \frac{R_g\, \omega_p\, (1 - S_C)}{R_b \cos\beta\, (1 + \sin\theta)}, \quad V_C = S_C\, R_g\, \omega_p, \tag{1}$$

where $\omega_c$ is shell circulation, $\omega_b$ is self-spin, $V_C$ is head-contact sliding speed, and $\cos\beta = R_g/\sqrt{R_g^2 + R_b^2 \sin^2\theta}$ it is set by geometry. We define

$$V_{\text{eff}}(\omega_p, S_C) = \sqrt{(R_g\omega_c)^2 + (R_b\omega_b)^2 + V_C^2}, \qquad R_T(S_C) \triangleq \frac{\omega_c}{\omega_p} = \frac{1 + S_C \sin\theta}{1 + \sin\theta}. \tag{2}$$

Our apparatus has $\theta$=45° and speeds $\{21, 48, 78, 108\}$ RPM. Real clips use shells bonded to the baseplate, so the head contact is nominally pure sliding ($S_C$=1) [19]; compliant-pad deformation, pressure gradients, and fluid effects can lower the effective slip [17], but per-clip $S_C$ is unobservable from overhead video. We therefore use $S_C \sim \text{Beta}(199.5, 10.5)$ (mean 0.95), rescaled to $[0.92, 0.98]$, as an uncertainty prior and regularizer, not a measured label [9]. To test slip when labels exist, we also build synthetic clips with $S_C \in \{0.20, 0.40, 0.60, 0.80\}$ (Sec. 6).

## 4 Method

The framework has three parts: a frozen video backbone with LoRA adapters, Fourier physics tokens injected by cross attention, and latent losses computed from differentiable kinematics.

**Backbone.** We adopt SVD-xt [4], which operates on VAE latents $\mathbf{z} \in \mathbb{R}^{T\times C\times h\times w}$ with v-prediction $\mathbf{v}_\theta = \sqrt{\bar\alpha_t}\boldsymbol{\epsilon} - \sqrt{1-\bar\alpha_t}\,\mathbf{x}_0$ and clean-latent recovery $\hat{\mathbf{x}}_0 = \sqrt{\bar\alpha_t}\,\mathbf{z}_t - \sqrt{1-\bar\alpha_t}\,\mathbf{v}_\theta$. We freeze the backbone, VAE, and CLIP encoder, and train rank-32 LoRA adapters [14] on attention projections with Min-SNR-$\gamma$ weighting ($\gamma$=5) [10].

**Fourier physics tokens.** RPM is normalized to $\widehat{\omega_p} = (\omega_p - \omega_{p,\min})/(\omega_{p,\max} - \omega_{p,\min}) \in [0, 1]$ over $\{21, 48, 78, 108\}$ RPM; $S_C$ is already fractional. Each scalar $s$ is lifted to $\gamma(s) = \big[\sin(2^k\pi s), \cos(2^k\pi s)\big]_{k=0}^{K-1}$ and encoded by an MLP into $\mathbb{R}^d$ ($d$=1024). The continuity of $\gamma$ supports interpolation to unseen speeds, tested by leave-one-speed-out evaluation (Sec. 6); it also permits off-grid queries such as 55 RPM, though we only validate held-out grid speeds here. Tokens $\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C}$ are combined by multi-head attention with the mean-token query, and the conditioning sequence is

$$\mathbf{T}_{\text{int}} = \text{LN}\big(\bar{\mathbf{T}} + \text{MHA}(\bar{\mathbf{T}}, \mathbf{T}_{12}, \mathbf{T}_{12})\big), \quad \mathbf{E} = [\mathbf{e}_{\text{img}}; \mathbf{T}_{\text{rpm}}; \mathbf{T}_{S_C}; \mathbf{T}_{\text{int}}] \in \mathbb{R}^{4\times d}, \tag{3}$$

with $\mathbf{T}_{12} = [\mathbf{T}_{\text{rpm}}; \mathbf{T}_{S_C}]$ and $\bar{\mathbf{T}} = \frac{1}{2}(\mathbf{T}_{\text{rpm}} + \mathbf{T}_{S_C})$. $\mathbf{E}$ enters every UNet cross-attention layer through LoRA-adapted $K, V$ projections; $Q$ stays frozen to reduce query drift. Self-attention $Q, K, V$ and output projections also receive LoRA adapters ($r$=$\alpha$=32) across temporal and spatial attention. Image and physics tokens are jointly dropped with $p_{\text{cfg}}$=0.1 for classifier-free guidance [12]. Encoding and mechanism ablations are in Sec. 6: FiLM replaces cross attention with a feature-wise affine modulation of the UNet activations, and raw $S_C$ keeps cross attention but removes the Fourier lift $\gamma$ for $S_C$, feeding the scalar directly.

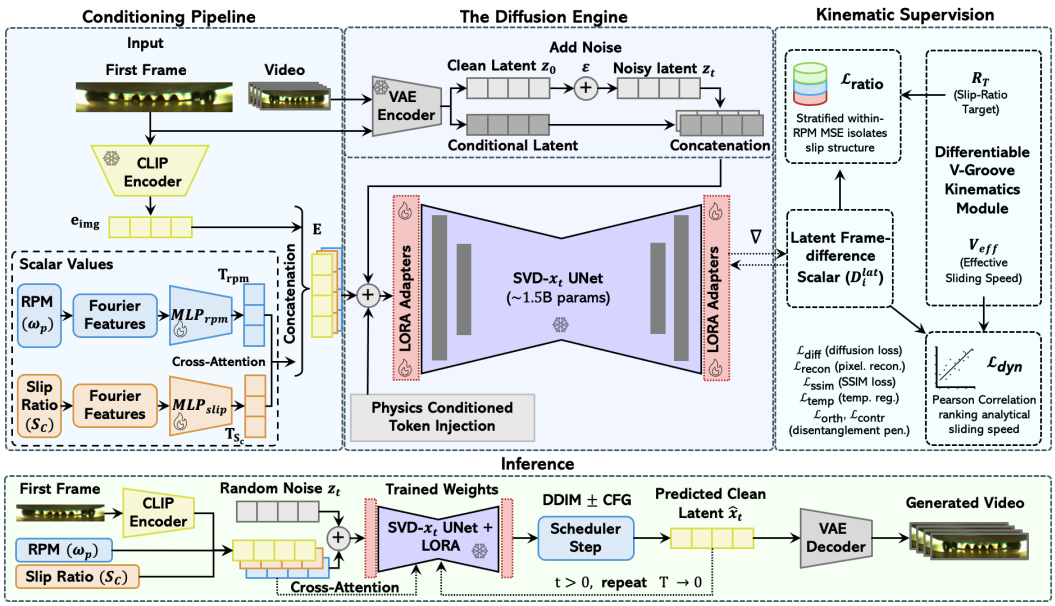



Figure 2: Method overview. SVD-xt is frozen (*gray*); only LoRA adapters and the Fourier physics encoder (*blue*) are trained. The operating scalars $(\omega_p, S_C)$ become cross-attention tokens and kinematic targets $V_{\text{eff}}, R_T$ for latent losses $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$ on $\hat{\mathbf{x}}_0$. At inference (bottom row) there is no target clip: starting from random noise, the same frozen encoders and trained adapters are reused to alternate the UNet with the sampling scheduler, and the frozen VAE decodes only the final latent to pixels.

**Differentiable kinematics module.** Equations (1)–(2) map physical $(\omega_p, S_C)$ to $V_{\text{eff}}, \omega_c, \omega_b, R_T$. These are scalar targets only: gradients enter LoRA through the latent frame-difference scalar $D^{\text{lat}}$ computed on $\hat{\mathbf{x}}_0$, not through the kinematic constants. Kinematics use physical RPM; only tokens use normalized RPM.

**Dynamics consistency loss $\mathcal{L}_{\text{dyn}}$.** From $\hat{\mathbf{x}}_0$ we form a per-clip latent frame-difference scalar

$$D_i^{\text{lat}} = \frac{1}{(T-1)Chw} \sum_{t=1}^{T-1} \left\| \hat{\mathbf{x}}_{0,i}^{(t+1)} - \hat{\mathbf{x}}_{0,i}^{(t)} \right\|_1, \qquad \mathcal{L}_{\text{dyn}} = -\operatorname{corr}\left(V_{\text{eff}}^{(1:B)}, D^{\text{lat}\,(1:B)}\right). \tag{4}$$

Unlike PDE-residual losses [3] that constrain each sample, $\mathcal{L}_{\text{dyn}}$ enforces a *ranking* across samples consistent with the analytical sliding speed.

**Stratified velocity-ratio loss $\mathcal{L}_{\text{ratio}}$.** Naively replacing $V_{\text{eff}}$ with $R_T(S_C)$ in Eq. (4) fails because between-RPM dynamics dominate, while the within-RPM variance of $R_T(S_C)$ under the real Beta prior is only $\operatorname{Var}(R_T \mid r) \approx 3.85 \times 10^{-5}$. A pooled correlation, therefore, sees mostly RPM. so we stratify by RPM. This also explains why the loss is a weak signal at $B \approx 8$–$16$: it regularizes ordinal structure rather than providing dense per-sample supervision. For each stratum $\mathcal{B}_r$ in the batch, define mean-normalized signals $\tilde{D}^{(i)} = D_i^{\text{lat}} / \operatorname{sg}\left[\overline{D^{\text{lat}}}_{\mathcal{B}_r}\right]$, $\tilde{R_T}^{(i)} = R_T(S_C^{(i)}) / \overline{R_T}_{\mathcal{B}_r}$, where $\overline{D^{\text{lat}}}_{\mathcal{B}_r} = \frac{1}{|\mathcal{B}_r|} \sum_{j \in \mathcal{B}_r} D_j^{\text{lat}}$ and analogously for $\overline{R_T}_{\mathcal{B}_r}$. The stop-gradient $\operatorname{sg}[\cdot]$ (`.detach()`) on the stratum-mean denominator prevents the loss from collapsing the predicted ratios to a degenerate scale. We use

$$\mathcal{L}_{\text{ratio}} = \frac{1}{|\mathcal{R}_B|} \sum_{r \in \mathcal{R}_B} \frac{1}{|\mathcal{B}_r|} \sum_{i \in \mathcal{B}_r} \left(\tilde{D}^{(i)} - \tilde{R_T}^{(i)}\right)^2, \tag{5}$$

including only strata with $|\mathcal{B}_r| \geq 2$ and within-stratum $\operatorname{Var}(R_T) > \varepsilon$. Mean-normalization removes RPM scale; MSE keeps a smooth gradient at $|\mathcal{B}_r| = 2$, where Pearson is degenerate at $\pm 1$.

**Role of the Beta prior on $S_C$.** In the real regime, $S_C$ is not measured per clip. We sample $S_C \sim \text{Beta}(199.5, 10.5)$ rescaled to $[0.92, 0.98]$ and use the sample as the conditioning token input and as the scalar argument to the kinematic targets $(V_{\text{eff}}, R_T)$. The prior has two limited roles: it

keeps the $S_C$ token in a plausible range and supplies weak within-stratum order for $\mathcal{L}_{\text{ratio}}$. Both physics losses are ranking objectives, not absolute slip regression; even a $\pm 0.03$ slip perturbation changes $R_T$ by only about 1.2 percentage points at $\theta$=45°. The Beta prior is therefore a regularizer, not a supervision oracle, and real-slip claims do not rely on it as ground truth. A wider prior would strengthen $\mathcal{L}_{\text{ratio}}$ gradients but would feed real clips slip values unsupported by the apparatus; a narrower prior would be physically safer but would nearly remove the slip signal. We choose the conservative middle and evaluate true slip only in $S$.

**Auxiliary objectives and a two-phase schedule.** A latent reconstruction loss $\mathcal{L}_{\text{recon}} = \ell_1(\hat{\mathbf{x}}_0, \mathbf{x}_0)$, $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{\mathbf{x}}_0, \mathbf{x}_0)$, $\mathcal{L}_{\text{temp}}$, and small token penalties $\mathcal{L}_{\text{orth}}, \mathcal{L}_{\text{contr}}$ (Appendix A4.2) complement $\mathcal{L}_{\text{diff}}$. Because physics losses need coherent $\hat{\mathbf{x}}_0$, Phase 1 trains perceptual losses only, and Phase 2 ramps physics terms with $w(e)$.

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_r \mathcal{L}_{\text{recon}} + \lambda_s \mathcal{L}_{\text{SSIM}} + w(e)\big[\lambda_d \mathcal{L}_{\text{dyn}} + \lambda_t \mathcal{L}_{\text{temp}} + \lambda_\rho \mathcal{L}_{\text{ratio}} + \lambda_o \mathcal{L}_{\text{orth}} + \lambda_c \mathcal{L}_{\text{contr}}\big]. \quad (6)$$

Defaults are reported in Appendix A4.

**Inference.** At test time, 25-step DDIM [28] with guidance scale $s$=3.0 generates a $T$=30-frame clip from a first frame and target $(\omega_p^\star, S_C^\star)$.

# 5 Theoretical Insights

The losses use only scalar kinematics, but their batch form needs three checks: $\mathcal{L}_{\text{dyn}}$ must be a valid correlation objective, $\mathcal{L}_{\text{ratio}}$ must expose the weak slip axis, and $D^{\text{lat}}$ must preserve visible-motion ordering. Full derivations are in Appendix A2.

## 5.1 Batch reliability of $\mathcal{L}_{\text{dyn}}$

$\mathcal{L}_{\text{dyn}}$ is the negative batch Pearson correlation between $V_{\text{eff}}(\omega_p, S_C)$ and $D^{\text{lat}}$ (Eq. 4). It is well posed when both have nonzero batch variance. RPM-stratified sampling supplies distinct speeds, and the two-phase schedule waits until reconstructed latents have temporal motion.

**Proposition 1** (Well-posed batch correlation)**.** *For $B \geq 2$, if $s^2_{B,V_{\text{eff}}} > 0$ and $s^2_{B,D^{\text{lat}}} > 0$, the batch Pearson correlation in $\mathcal{L}_{\text{dyn}}$ is defined, bounded in $[-1, 1]$, locally Lipschitz in the denoiser parameters away from the zero-variance set, and differentiable almost everywhere. For $B > 3$, under a bivariate normal reference model with population correlation $\rho$,*

$$\sqrt{B-3}\left(z_F(\hat{r}_B) - z_F(\rho)\right) \xrightarrow{d} \mathcal{N}(0,1), \quad z_F(r) = \tfrac{1}{2}\log\tfrac{1+r}{1-r}. \quad (7)$$

*Thus, under this asymptotic reference approximation, a Fisher-$z$ half-width $\varepsilon$ at confidence $1-\delta$ corresponds to $B \approx 3 + z^2_{1-\delta/2}/\varepsilon^2$.*

At $\varepsilon$=0.1, $\delta$=0.05, $B^*$=388. Our effective batches (8 on most real runs, 16 on synthetic LOSO) therefore use $\mathcal{L}_{\text{dyn}}$ as a noisy ordering signal, not a precise population estimate. Figure 3(a,b) shows the Fisher uncertainty and nonzero denominators under stratified batching.

## 5.2 Why $\mathcal{L}_{\text{ratio}}$ is stratified

$R_T(S_C)$ is affine in $S_C$ with slope $m = \sin\theta/(1+\sin\theta) \approx 0.414$ at $\theta$=45°. Under the real Beta prior, $\text{Var}(R_T \mid r) \approx 3.86 \times 10^{-5}$ is within an RPM.tratum, whereas $V_{\text{eff}}$ changes by about 5× from 21 to 108 RPM. A pooled slip correlation is therefore speed-dominated. Eq. (5) removes the RPM component, mean-normalizes strata, and uses MSE, so two-sample strata still have smooth gradients.

## 5.3 Latent and pixel motion

The physics losses use $D^{\text{lat}}$ instead of optical flow or pixel differences. On 120 real clips (Fig. 3(c)), $D^{\text{lat}}$ has pooled Pearson $r$=0.91 with pixel frame difference, per-RPM correlations $\{0.78, 0.83, 0.94, 0.92\}$, and low-band/high-band correlations $0.985/0.730$. Thus, it captures circulation-like visible motion more strongly than self-rotation; we treat this as a scope limit, not a contact-mechanics guarantee.

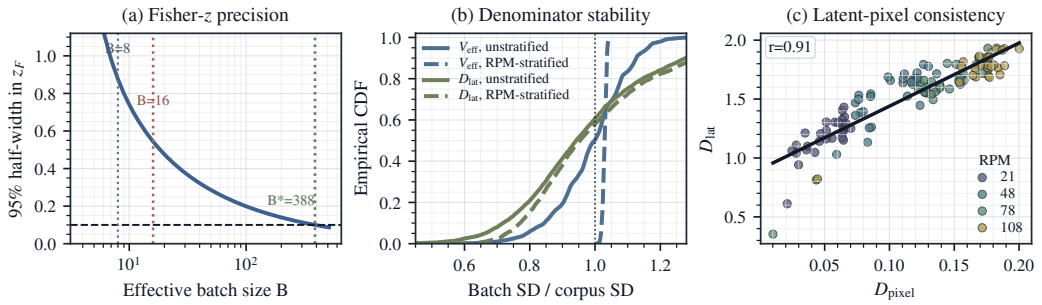



Figure 3: Foundations of the physics losses. **(a)** Fisher-$z$ uncertainty shows that $\mathcal{L}_{\text{dyn}}$ is a noisy batch signal at our compute budget. **(b)** RPM-stratified batching keeps the $\mathcal{L}_{\text{dyn}}$ denominator away from collapse. **(c)** The latent motion scalar $D^{\text{lat}}$ tracks pixel motion on real clips at $r{=}0.91$.

**Scope.** These arguments support scalar ordering objectives only. Generated videos are not claimed to reproduce contact mechanics, material removal, or roughness evolution; the training signal is a cross-clip kinematic prior, not a pixel-level physical law.

## 6 Experiments

$S_C \in [0, 1]$ denotes head-contact slip, from pure rolling (0) to pure sliding (1) [19]. Because real $S_C$ is unmeasured, regime $R$ evaluates realism and speed control under a narrow prior, while synthetic regime $S$ tests slip recovery with known $S_C$.

### 6.1 Datasets and evaluation

**Real regime** $R$**.** Four videos at $\{21, 48, 78, 108\}$ RPM yield 604 clips of $T{=}30$ frames (stride 5), split $70/20/10$ by source segment so overlapping windows do not cross splits. Each clip receives a persisted draw $S_C \sim \text{Beta}(199.5, 10.5)$ over $[0.92, 0.98]$ (Sec. 3); this regime grounds speed and realism but has no usable slip label.

**Synthetic regime** $S$**.** A renderer driven by Eqs. (1)–(2) produces matched clips at the same RPMs and known slips $\{0.20, 0.40, 0.60, 0.80\}$, supplying the wide slip variation absent from $R$. It is a positive-control renderer rather than a claim of photorealistic polishing: it matches the orbit geometry and timing used by the scorer but only approximates real lighting, fluid, specularities, and partial occlusions. We therefore use $S$ to test identifiability under known labels, not to claim synthetic-to-real slip transfer. Because renderer and scorer share kinematic geometry, success on $S$ is best read as a controlled observability test with known slip labels, not an independent physics audit; the real-domain conclusion still comes from $R$ under the current evaluation setup.

**Metrics.** Visual metrics are paired PSNR, SSIM, LPIPS, FID-I3D, and FVD [11, 32]; FID-I3D uses the FVD RGB-I3D Kinetics-400 backbone, so comparisons are relative. Control uses a $4 \times 4 \times 4$ sweep per seed (scene $\times$ RPM $\times$ slip): speed rank orders $\widehat{\omega_c}$ across RPMs at fixed scene/slip, and slip rank orders $\widehat{R_T}$ across slips at fixed scene/RPM. Real sweeps use $S_C \in \{0.92, 0.94, 0.96, 0.98\}$; synthetic sweeps use the labeled $S$ grid. Results average five seeds unless a deterministic baseline is identified.

**Tracking and inversion.** Scoring uses fine-tuned YOLO11n detections stitched into tracks, circulation recovery $\widehat{\omega_c}$, and Eq. (1) inversion (Fig. 4). It is independent of training: losses use latent $D^{\text{lat}}$ and scalar $V_{\text{eff}}(\omega_p, S_C)$, while scoring uses only pixels, tracking gates (length, singularity, $R^2$, cross-track MAD, sign, aggregation, plausibility), and a validation-fitted calibration $c_{\text{suite}}$ applied before inversion. This scalar is fit once per suite on calibration clips and then fixed across variants, seeds, and test sweeps. No test labels, $D^{\text{lat}}$, or loss values enter scoring. Yield is the fraction of plausible sweep cells; ranks or correlations use passing cells only and are reported with yield where attrition is material. Sweeping $r_{t,\min} \in \{0.30, \ldots, 0.90\}$ with an upper bound 1.35 keeps slip rank in 0.42–0.54 (mean 0.51), so rank order is not threshold-driven. Details are in Appendix A3.4. Because the same analytical kinematics define targets and inversion, the evaluation can still reward videos

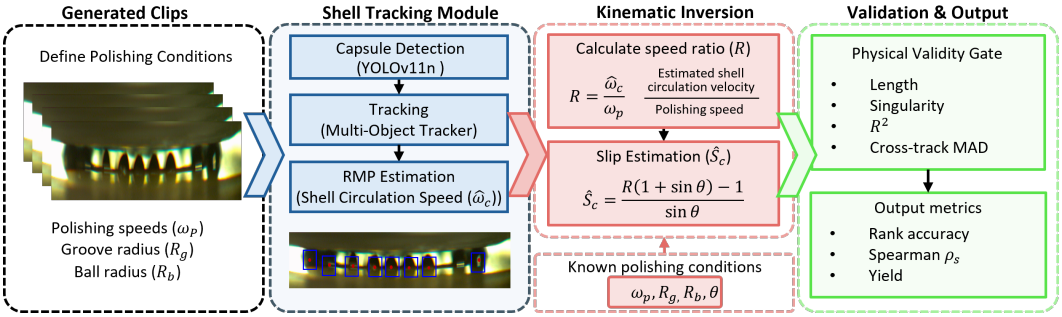



Figure 4: Slip evaluation. Generated clips pass through capsule detection, tracking, circulation recovery, kinematic inversion, and a physical validity gate before rank accuracy is computed.

Table 1: Real design study ($R$), five-seed mean. Visual metrics use the paired test split; speed/slip use the gated sweep. Chance slip rank is 0.5; bold marks best within group; † marks default. Yield is the fraction of clips passing the physical validity gate.

| | | Visual | | | | | Speed | | Slip | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Group** | **Variant** | PSNR↑ | SSIM↑ | LPIPS↓ | FID-I3D↓ | FVD↓ | rank↑ | $\rho_s$ ↑ | rank [95% CI]↑ | $\rho_s$ ↑ | Yield |
| Loss | conditioning only | **17.73** | **0.710** | **0.157** | **159.2** | 193.5 | 0.981 | 0.913 | **0.512** [.38,.64] | **+0.024** | 0.956 |
| | + $\mathcal{L}_{\text{dyn}}$ | 16.58 | 0.688 | 0.185 | 162.5 | **192.7** | 0.975 | 0.912 | 0.508 [.39,.63] | +0.006 | 0.959 |
| | + $\mathcal{L}_{\text{ratio}}$ | 17.06 | 0.697 | 0.173 | 164.6 | 200.4 | 0.979 | **0.921** | **0.512** [.37,.65] | -0.003 | 0.975 |
| | + both ($\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$)† | 16.69 | 0.689 | 0.178 | 160.7 | 194.9 | **0.985** | 0.909 | 0.502 [.37,.63] | -0.033 | 0.959 |
| Cond. | FiLM | 16.49 | 0.680 | 0.180 | 171.7 | 204.6 | 0.977 | 0.909 | 0.492 [.37,.62] | -0.003 | 0.953 |
| | Raw $S_C$ (no Fourier) | 15.82 | 0.674 | 0.190 | 161.5 | 192.3 | 0.981 | 0.911 | **0.514** [.37,.66] | -0.013 | 0.934 |
| Schedule | Joint training | 16.58 | 0.687 | 0.178 | 163.9 | 193.7 | 0.973 | 0.911 | **0.527** [.40,.65] | **+0.037** | 0.959 |
| | Perceptual pretrain | **16.82** | **0.694** | **0.172** | 156.0 | 187.9 | **0.981** | **0.918** | 0.498 [.37,.63] | +0.012 | 0.944 |
| | Pretrain + phys. fine tune† | 16.75 | 0.690 | 0.176 | **151.6** | **176.2** | 0.975 | 0.906 | 0.500 [.37,.62] | -0.022 | 0.953 |

that produce the right centroid orbit without proving true self-spin or contact physics. We address this by keeping training and scoring signals disjoint, reporting gate yield, and treating rank scores as kinematic controllability rather than physical truth.

### 6.2 Real regime design study

Table 1 varies one factor at a time around the full model: cross attention, both physics losses, $K$=16, $B_{\text{eff}}$=8, and two-phase training. The full model is rerun per group, so small differences reflect five-seed variance.

Speed is robust in $R$: every noncollapsed variant reaches rank $\geq 0.97$. Slip stays at chance with CIs spanning 0.5, as expected for unmeasured $S_C$ restricted to $[0.92, 0.98]$. The "conditioning only" row removes $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$ but keeps RPM and $S_C$ tokens; its speed rank 0.981 shows that conditioning already steers speed, while physics losses give the best loss-group speed rank (0.985) and the best $S$ FID-I3D/FVD below. Conditioning and schedule mainly tune appearance. Supplementary sweeps show stable speed rank for $B_{\text{eff}} \in \{2, 4, 8, 16\}$ and $K \in \{4, 8, 16\}$, with collapse at $K$=32.

### 6.3 Synthetic slip identifiability

Regime $S$ tests slip identifiability with informative labels. Table 2 shows speed rank $\geq 0.99$ and slip rank $\geq 0.735$ for all settings, with CIs excluding chance.

Cross attention alone exploits the wide synthetic slip grid, so physics losses do not raise slip rank on $S$; they improve distributional quality for the combined model (FID-I3D 236.4, FVD 116.0). The "conditioning only" row is not image-only: it still receives Fourier RPM and slip tokens, and clean synthetic labels make the ordinal slip axis easy for conditioning alone. The losses optimize noisier surrogates: $\mathcal{L}_{\text{dyn}}$ is speed-dominated, while $\mathcal{L}_{\text{ratio}}$ depends on $D^{\text{lat}}$, which is more sensitive to total visible displacement than to slip-specific circulation or self-spin. With finite batches, this can trade a few rank points for better video-distribution metrics. Learned directional flow or video features may

Table 2: Synthetic loss sweep ($S$), five-seed mean, plausibility gate, known $S_C \in \{0.20, 0.40, 0.60, 0.80\}$. Bold marks best per column. Yield is the fraction of clips passing the physical validity gate.

| **Losses** | PSNR↑ | SSIM↑ | LPIPS↓ | FID-I3D↓ | FVD↓ | Speed rank↑ | Slip rank [95% CI]↑ | Slip $\rho_s$ ↑ | Yield |
|---|---|---|---|---|---|---|---|---|---|
| conditioning only | 21.39 | 0.733 | 0.203 | 243.7 | 126.9 | **0.996** | **0.769** [.63,.90] | **+0.615** | 0.678 |
| $+ \mathcal{L}_{\text{dyn}}$ | **21.61** | 0.719 | **0.197** | 246.1 | 127.9 | 0.992 | 0.736 [.59,.86] | +0.543 | 0.669 |
| $+ \mathcal{L}_{\text{ratio}}$ | 21.36 | 0.723 | 0.200 | 245.5 | 126.4 | **0.996** | 0.766 [.63,.88] | +0.550 | 0.688 |
| $+\mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{ratio}}$ | 21.58 | **0.734** | 0.201 | **236.4** | **116.0** | **0.996** | 0.735 [.59,.87] | +0.507 | 0.666 |

Table 3: Leave-one-speed-out generalization, five-seed mean, plausibility gate. $R$ has unobserved slip; $S$ has known $S_C \in \{0.20, 0.40, 0.60, 0.80\}$.

| | | **Visual** | | | | **Speed** | **Slip** | |
|---|---|---|---|---|---|---|---|---|
| **Held out** | **Type** | PSNR↑ | SSIM↑ | LPIPS↓ | FID-I3D↓ | rank [95% CI]↑ | rank [95% CI]↑ | Yield |
| *Regime R (real polishing; GT slip unobservable)* | | | | | | | | |
| 21 RPM | boundary | 11.66 | 0.540 | 0.315 | **147.1** | 0.631 [.52,.74] | 0.492 [.34,.64] | 0.884 |
| 48 RPM | interior | 11.81 | 0.547 | 0.307 | 144.4 | 0.827 [.75,.90] | 0.487 [.34,.64] | 0.822 |
| 78 RPM | interior | **12.78** | **0.581** | **0.279** | 175.0 | **0.875** [.81,.93] | **0.588** [.44,.73] | 0.922 |
| 108 RPM | boundary | 12.01 | 0.561 | 0.298 | 169.7 | 0.640 [.61,.67] | 0.485 [.36,.60] | **0.981** |
| *Regime S (synthetic; GT slip known)* | | | | | | | | |
| 21 RPM | boundary | 20.44 | **0.725** | 0.238 | **228.2** | 0.696 [.42,.92] | 0.680 [.44,.89] | 0.416 |
| 48 RPM | interior | 20.12 | 0.670 | 0.241 | 253.9 | **0.938** [.86,.99] | 0.722 [.56,.86] | 0.634 |
| 78 RPM | interior | 19.75 | 0.660 | 0.250 | 249.5 | 0.721 [.62,.82] | **0.742** [.62,.86] | 0.709 |
| 108 RPM | boundary | **20.48** | 0.719 | **0.235** | 233.0 | 0.665 [.65,.68] | 0.723 [.60,.83] | **0.884** |

help, but scalar flow magnitude is insufficient because it measures the amount of motion, not whether motion follows $R_T(S_C)$ (Appendix A3.2). Thus, slip is learnable when exposed; the real bottleneck is $S_C$ observability.

### 6.4 Leave one speed out

We hold out one RPM during training and evaluate the full model on the held-out fold. Interior folds (48, 78 RPM) test interpolation; boundary folds (21, 108 RPM) test extrapolation.

Both regimes show easier speed interpolation than boundary extrapolation. Slip diverges by regime: real LOSO remains at chance, while synthetic LOSO is above chance on all folds and excludes chance on three of four.

### 6.5 Non-generative flow baselines

Two training-free flow-warping baselines test whether simple kinematics can replace generation. **GT-match** warps the first real frame with sinusoidal displacement matched to ground-truth flow magnitude at each RPM, making slip flat by construction. **Analytical** uses $\omega_c = \omega_p(1+S_C \sin\theta)/(1+\sin\theta)$ from Eq. 1, activating slip tautologically. The baselines are deterministic sanity checks on $S_C \in \{0.30, 0.50, 0.70, 0.95\}$. To avoid mixing protocols, our model row reuses the five-seed real-regime full-model result from Table 1; real slip therefore remains at chance in both tables.

Our model has higher paired fidelity than flow baselines (PSNR 16.69 vs. 13.60/12.37), but real slip rank is chance (0.502, CI spanning 0.5), matching Table 1. The analytical ceiling (0.804) is tautological and inaccessible without prescribing the true orbit. Flow baselines win FID-I3D/FVD because they warp real frames; PSNR/SSIM/LPIPS favor the generator. The real-data takeaway is speed controllability and visual fidelity, not demonstrated slip controllability.

## 7 Discussion

The experiments separate a positive control from an identifiability limit. Speed control is reliable: noncollapsed $R$ variants reach rank $\geq 0.97$, and the same recipe reaches $\geq 0.99$ in $S$. A useful envelope is $K \in \{4, 8, 16\}$, cross attention, and effective batch sizes 4–16; the schedule mainly tunes FID-I3D/FVD. Slip control depends on labels: $S$ reaches rank $\geq 0.735$ with CIs excluding

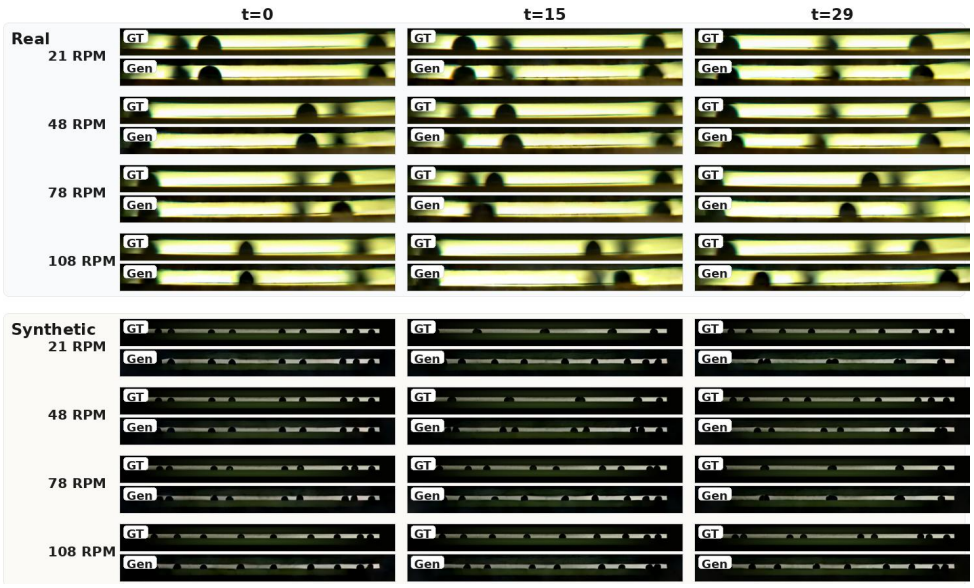



Figure 5: Representative RPM sweep for the full model. Displacement increases with RPM, consistent with the robust speed axis ranking in Table 1; slip effects are subtler and are evaluated through tracking and Lee inversion. Appendix A3.6 shows every frame of a generated clip at each speed for both regimes.

Table 4: Non-generative flow baselines and the five-seed real-regime model result. Baselines are deterministic wide-slip sanity checks; the model row is the same narrow-prior real result as Table 1, including its gate yield of 0.959.

| **Method** | PSNR↑ | SSIM↑ | LPIPS↓ | FID-I3D↓ | FVD↓ | Slip rank [95% CI]↑ |
|---|---|---|---|---|---|---|
| GT-match (flow) | 13.60 | 0.679 | 0.248 | **41.8** | 37.9 | 0.484 (at chance) |
| Analytical (kinematic) | 12.37 | 0.591 | 0.306 | 43.3 | **26.1** | 0.804 (tautological†) |
| **Ours (full model, 5 seeds)** | **16.69** | **0.689** | **0.178** | 160.7 | 194.9 | 0.502 [.37,.63] |

† The analytical baseline drives the warp with the same kinematic equation the scorer inverts, making its rank accuracy a tautological ceiling, not an achievable target.

chance, while $R$ stays at chance because $S_C$ is unmeasured and restricted to $[0.92, 0.98]$. We claim slip learnability under informative labels, not the transfer of synthetic slip control to real polishing.

Main limitations are a small number of video clips 604 from one real regime, fewer passing plausibility cells in $R$ than in $S$, weaker boundary LOSO generalization, and lower held-out-speed appearance. The surrogate also stops short of being a full process-optimization tool: it predicts plausible kinematic videos, not material removal, roughness, or defect formation. Its immediate use is exploratory design-of-experiments support, sanity-checking operating sweeps, and stress-testing downstream trackers; closed-loop polishing decisions would require independent metrology linking generated kinematics to surface outcomes. Boundary LOSO failures are consistent with extrapolating both appearance and speed statistics beyond the observed range; interior folds mostly test interpolation and are more stable. This framing keeps the practical claim modest under current sensing constraints and data: the model is useful for asking whether a requested operating change produces plausible video kinematics, not for certifying a polishing recipe by itself.

**Self-spin $\omega_b$ and unobservable DOF.** The kinematic model predicts orbit ($\omega_c$) and self-spin ($\omega_b$). Our centroid tracker recovers $\omega_c$, but $\omega_b$ appears mainly as surface texture and needs multi-angle cameras or model-based texture analysis. Since $D^{\text{lat}}$ tracks pixel motion ($r$=0.91) but emphasizes circulation, the model is a kinematic video surrogate, not a contact-mechanics or material-removal simulator.

$K{=}32$ **collapse.** With only four training RPMs, $K{=}32$ introduces harmonics at $k \geq 16$ that exceed the four-point Nyquist rate, encouraging grid overfit and collapsed speed rank. $K{=}16$ is the recommended default.

**Compute and scalability.** Each run costs 12–16 GPU-hours on one RTX 4090; inference takes 35–45 s per 30-frame $496{\times}48$ clip. Rank-32 LoRA adds $\approx$ 22M trainable parameters to frozen 1.5B-parameter SVD-xt. Larger slip grids or geometries need more data, but only the kinematics must be re-parameterized.

**Future directions.** An EM loop alternating latent slip estimation with model updates may reduce the need for per-clip slip measurement; $S$ shows the generator can recover slip when labels exist. Learned keypoints could recover both $\omega_c$ and $\omega_b$, and the renderer enables virtual sweeps over groove angle, capsule size, and speed before physical trials.

**Broader impacts.** The work targets manufacturing video surrogates for ICF capsule polishing, where controllable generation may reduce process tuning costs. It does not model implosion physics or add information beyond public polishing kinematics, but ICF has dual-use exposure and release plans that should be reviewed accordingly. The main misuse risk is treating generated clips as physical evidence beyond the scope; we separate perceptual and kinematic metrics, state that real slip is unmeasured, and require independent validation before closed-loop use.

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## References

[1] H Abu-Shawareb, R Acree, P Adams, J Adams, B Addis, R Aden, P Adrian, BB Afeyan, M Aggleton, L Aghaian, et al. Achievement of target gain larger than unity in an inertial fusion experiment. *Physical review letters*, 132(6):065102, 2024.

[2] Rick Akkerman, Haiwen Feng, Michael J Black, Dimitrios Tzionas, and Victoria Fernández Abrevaya. Interdyn: Controllable interactive dynamics with video diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12467–12479, 2025.

[3] Jan-Hendrik Bastek, WaiChing Sun, and Dennis Kochmann. Physics-informed diffusion models. In *The Thirteenth International Conference on Learning Representations*, 2025.

[4] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.

[5] Boyuan Chen, Hanxiao Jiang, Shaowei Liu, Saurabh Gupta, Yunzhu Li, Hao Zhao, and Shenlong Wang. Physgen3d: Crafting a miniature interactive world from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6178–6189, 2025.

[6] Minkyoung Cho, Ruben Ohana, Christian Jacobsen, Adityan Jothi, Min-Hung Chen, Z Morley Mao, and Ethem Can. Tc-lora: Temporally modulated conditional lora for adaptive diffusion control. *arXiv preprint arXiv:2510.09561*, 2025.

[7] Gunnar Farnebäck. Two-frame motion estimation based on polynomial expansion. In *Scandinavian conference on Image analysis*, pages 363–370. Springer, 2003.

[8] Ronald A Fisher. On the" probable error" of a coefficient of correlation deduced from a small sample. *Metron*, 1:3–32, 1921.

[9] Andrew Gelman, John B Carlin, Hal S Stern, and Donald B Rubin. *Bayesian data analysis*. Chapman and Hall/CRC, 1995.

[10] Tiankai Hang, Shuyang Gu, Chen Li, Jianmin Bao, Dong Chen, Han Hu, Xin Geng, and Baining Guo. Efficient diffusion training via min-snr weighting strategy. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7441–7451, 2023.

[11] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.

[12] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[13] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *Advances in neural information processing systems*, 35:8633–8646, 2022.

[14] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022.

[15] Jiahe Huang, Guandao Yang, Zichen Wang, and Jeong Joon Park. Diffusionpde: Generative pde-solving under partial observation. *Advances in Neural Information Processing Systems*, 37:130291–130323, 2024.

[16] Omar A Hurricane, PK Patel, R Betti, Dustin H Froula, SP Regan, Stephen A Slutz, MR Gomez, and Mary Ann Sweeney. Physics principles of inertial confinement fusion and us program overview. *Reviews of Modern Physics*, 95(2):025005, 2023.

[17] Robert L Jackson, Hamed Ghaednia, Hyeon Lee, Amir Rostami, and Xianzhang Wang. Contact mechanics. *Tribology for scientists and engineers: from basics to advanced concepts*, pages 93–140, 2013.

[18] Maurice G Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938.

[19] Rong-Tsong Lee, Yih-Chyun Hwang, and Yuang-Cherng Chiou. Lapping of ultra-precision ball surfaces. part i. concentric v-groove lapping system. *International Journal of Machine Tools and Manufacture*, 46(10):1146–1156, 2006.

[20] Rong-Tsong Lee, Yih-Chyun Hwang, and Yuang-Cherng Chiou. Lapping of ultra-precision ball surfaces. part ii. eccentric v-groove lapping system. *International Journal of Machine Tools and Manufacture*, 46(10):1157–1169, 2006.

[21] Shaowei Liu, Zhongzheng Ren, Saurabh Gupta, and Shenlong Wang. Physgen: Rigid-body physics-grounded image-to-video generation. In *European Conference on Computer Vision*, pages 360–378. Springer, 2024.

[22] Antonio Montanaro, Luca Savant Aira, Emanuele Aiello, Diego Valsesia, and Enrico Magli. Motioncraft: Physics-based zero-shot video generation. *Advances in Neural Information Processing Systems*, 37:123155–123181, 2024.

[23] EI Moses, RN Boyd, BA Remington, CJ Keane, and R Al-Ayat. The national ignition facility: Ushering in a new age for high energy density science. *Physics of Plasmas*, 16(4), 2009.

[24] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4195–4205, 2023.

[25] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[26] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

[27] Peter J Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical association*, 88(424):1273–1283, 1993.

[28] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.

[29] Matthew Tancik, Pratul Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in neural information processing systems*, 33:7537–7547, 2020.

[30] Zachary Teed and Jia Deng. Raft: Recurrent all-pairs field transforms for optical flow. In *European conference on computer vision*, pages 402–419. Springer, 2020.

[31] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078–10093, 2022.

[32] Thomas Unterthiner, Sjoerd Van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.

[33] Aad W Van der Vaart. *Asymptotic statistics*, volume 3. Cambridge university press, 2000.

[34] Chen Wang, Chuhao Chen, Yiming Huang, Zhiyang Dou, Yuan Liu, Jiatao Gu, and Lingjie Liu. Physctrl: Generative physics for controllable and physics-grounded video generation. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026.

[35] Zhouxia Wang, Ziyang Yuan, Xintao Wang, Yaowei Li, Tianshui Chen, Menghan Xia, Ping Luo, and Ying Shan. Motionctrl: A unified and flexible motion controller for video generation. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–11, 2024.

[36] Zexi Wu, Baolu Li, Jing Dai, Yiming Zhang, Yue Ma, Qinghe Wang, Xu Jia, and Hongming Xu. Video2lora: Unified semantic-controlled video generation via per-reference-video lora. *arXiv preprint arXiv:2603.08210*, 2026.

[37] Xindi Yang, Baolu Li, Yiming Zhang, Zhenfei Yin, Lei Bai, Liqian Ma, Zhiyong Wang, Jianfei Cai, Tien-Tsin Wong, Huchuan Lu, et al. Vlipp: Towards physically plausible video generation with vision and language informed physical prior. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 12360–12370, 2025.

[38] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models. *arXiv preprint arXiv:2308.06721*, 2023.

[39] Ye Yuan, Jiaming Song, Umar Iqbal, Arash Vahdat, and Jan Kautz. Physdiff: Physics-guided human motion diffusion model. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 16010–16021, 2023.

[40] Ke Zhang, Cihan Xiao, Jiacong Xu, Yiqun Mei, and Vishal M Patel. Think before you diffuse: Infusing physical rules into video diffusion. *arXiv preprint arXiv:2505.21653*, 2025.

[41] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3836–3847, 2023.

[42] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018.

[43] Tianyuan Zhang, Hong-Xing Yu, Rundi Wu, Brandon Y Feng, Changxi Zheng, Noah Snavely, Jiajun Wu, and William T Freeman. Physdreamer: Physics-based interaction with 3d objects via video generation. In *European Conference on Computer Vision*, pages 388–406. Springer, 2024.

[44] Yifu Zhang, Peize Sun, Yi Jiang, Dongdong Yu, Fucheng Weng, Zehuan Yuan, Ping Luo, Wenyu Liu, and Xinggang Wang. Bytetrack: Multi-object tracking by associating every detection box. In *European conference on computer vision*, pages 1–21. Springer, 2022.

[45] Zhenghao Zhang, Junchao Liao, Menghao Li, Zuozhuo Dai, Bingxue Qiu, Siyu Zhu, Long Qin, and Weizhi Wang. Tora: Trajectory-oriented diffusion transformer for video generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 2063–2073, 2025.

# Appendix

## A1 Kinematic Equations

This supplementary section derives Eqs. (1)–(2) from first principles, following Lee et al. [19, 20]. The analysis assumes a rigid ball of radius $R_b$ resting in a concentric V-groove of half-angle $\theta$ and groove radius $R_g$ cut into a baseplate spinning at $\omega_p$, with an upper lap loaded against the ball. Contact $A$ is the inner groove wall, $B$ the outer groove wall, and $C$ is the head contact with the upper lap.

### A1.1 Geometry

The ball contacts $A$ and $B$ at radial distances

$$R_i \;=\; R_g - R_b\cos\theta, \qquad R_o \;=\; R_g + R_b\cos\theta, \tag{8}$$

from the baseplate axis, and $C$ at a radial distance $R_g$. The ball's spin axis tilts by $\beta$ with $\tan\beta = R_b\sin\theta/R_g$; $\beta$ depends only on geometry.

### A1.2 No-slip at the groove walls

Let $\omega_c$ be the circulation angular speed about the baseplate axis and $\omega_b$ the self-spin. Linear contact velocities at $A$ and $B$ from baseplate motion must be balanced by ball-surface velocities:

$$R_i\omega_p = R_i\omega_c + R_b\omega_b\sin(\theta-\beta), \qquad R_o\omega_p = R_o\omega_c + R_b\omega_b\sin(\theta+\beta). \tag{9}$$

These are the no-slip conditions at $A$ and $B$.

### A1.3 Slip at the head contact

At contact $C$,

$$V_C \;=\; R_g\,\omega_{c,\mathrm{head}} - R_g\,\omega_c \;=\; S_C\,R_g\,\omega_p, \tag{10}$$

where $\omega_{c,\mathrm{head}}$ is the no-slip ball-circulation speed and $S_C \in [0,1]$ is the dimensionless head-contact slip ratio. Combining Eq. (8)–(10) and using standard trigonometric identities yields

$$\omega_c = \omega_p\,\frac{1+S_C\sin\theta}{1+\sin\theta}, \qquad \omega_b = \frac{R_g\,\omega_p\,(1-S_C)}{R_b\cos\beta\,(1+\sin\theta)}, \tag{11}$$

which, together with Eq. (10), gives the three components of Eq. (1).

### A1.4 Effective sliding speed and slip-ratio target

Aggregating the three component *surface speeds*, $V_{\text{eff}} \triangleq \sqrt{(R_g\omega_c)^2 + (R_b\omega_b)^2 + V_C^2}$ is the first half of Eq. (2). Multiplying the angular velocities by their physical radii keeps all terms in m/s, so the target is invariant to the choice of length unit. Since each component speed scales linearly with $\omega_p$ for fixed $S_C$, $V_{\text{eff}}$ is linear in $\omega_p$ with zero intercept and is strictly increasing in $\omega_p$, which is the linearity used by $\mathcal{L}_{\text{dyn}}$ (Eq. 4). The circulation-to-plate ratio $R_T(S_C) = \omega_c/\omega_p = (1 + S_C \sin\theta)/(1 + \sin\theta)$ is strictly increasing in $S_C$ for $\theta \in (0, \pi/2)$, providing the analytical ordinal target used by $\mathcal{L}_{\text{ratio}}$ (Eq. 5).

## A2 Theory Derivations

This supplementary section backs Sec. 5: (i) well-posedness and asymptotic calibration of $\mathcal{L}_{\text{dyn}}$ (Sec. 5.1); (ii) the variance-decomposition argument for why $\mathcal{L}_{\text{ratio}}$ is stratified (Sec. 5.2); and (iii) the latent–pixel consistency diagnostic, with the empirical instantiation $r$=0.91 on $n$=120 clips (Sec. 5.3).

### A2.1 Well-posedness and asymptotic calibration of $\mathcal{L}_{\text{dyn}}$

**Setup.** Write $V_i \triangleq V_{\text{eff}}(\omega_p^{(i)}, S_C^{(i)})$ and $D_i \triangleq D_i^{\text{lat}}$ (Eq. 4). The finite-batch non-degeneracy hypotheses are $s_{B,V}^2 > 0$ and $s_{B,D}^2 > 0$, where $s_{B,V}^2 \triangleq (B-1)^{-1}\sum_i (V_i - \bar{V})^2$ and analogously for $D$. A sufficient practical condition for $s_{B,V}^2 > 0$ is that the batch contain at least two realized speeds $V_{\text{eff}}$; our stratified sampler enforces this by placing multiple physical RPM strata in each batch.

**Well-posedness.** The batch-Pearson correlation is

$$\hat{r}_B \;=\; \frac{\sum_{i=1}^{B}(V_i - \bar{V})(D_i - \bar{D})}{\sqrt{\sum_i (V_i - \bar{V})^2}\,\sqrt{\sum_i (D_i - \bar{D})^2}}. \tag{12}$$

The empirical variances $\frac{1}{B}\sum_i (V_i - \bar{V})^2$ and $\frac{1}{B}\sum_i (D_i - \bar{D})^2$ are strictly positive for $B \geq 2$ under the hypotheses above, so the denominator is positive and $\hat{r}_B \in [-1, 1]$ is well-defined. $\mathcal{L}_{\text{dyn}} = -\hat{r}_B$ inherits well-posedness. The numerator and denominator are locally Lipschitz functions of $\{D_i\}$ away from the zero-variance set. If the denoiser outputs are locally Lipschitz functions of the parameters, the composition is locally Lipschitz in the denoiser parameters away from the zero-variance set. Because each $D_i$ is an $\ell_1$ latent frame-difference, differentiability is almost everywhere: the only additional excluded set is where an adjacent latent pixel difference is exactly zero.

**Asymptotic calibration via Fisher's $z$-transform.** For bivariate-normal $(V, D)$ with correlation $\rho \in (-1, 1)$, Fisher's $z$-transform $z_F(r) = \frac{1}{2}\log\frac{1+r}{1-r} = \tanh^{-1}(r)$ satisfies, classically [see e.g. 8],

$$\sqrt{B-3}\,\big(z_F(\hat{r}_B) - z_F(\rho)\big) \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } B \to \infty. \tag{13}$$

An asymptotic $(1-\delta)$-CI on the $z_F$-scale therefore has a half-width $z_{1-\delta/2}/\sqrt{B-3}$.

Setting this asymptotic half-width to be at most $\varepsilon$ corresponds to the reference size $B^* = \left\lceil 3 + z_{1-\delta/2}^2/\varepsilon^2 \right\rceil$. For $\varepsilon$=0.1, $\delta$=0.05, $z_{1-\delta/2}$=1.96, so $B^* = \lceil 3 + (1.96)^2/0.01 \rceil = 388$.

With only finite fourth moments, the sample covariance vector is still asymptotically normal by the multivariate CLT and $\hat{r}_B$ follows by the delta method [33], but the variance is distribution-dependent. Thus the $1/(B-3)$ formula is a Gaussian reference calculation, not an exact finite-batch interval for our four-RPM grid.

**Stratified-batching corollary.** If $\{\omega_p^{(i)}\}_{i=1}^B$ is constructed so each batch contains at least one sample in each of a fixed set of $\omega_p$-strata, the empirical variance $\frac{1}{B}\sum_i (V_i - \bar{V})^2$ is bounded below by the between-strata variance of $V_{\text{eff}}$, a deterministic constant $\sigma_{\min}^2 > 0$. This tightens the denominator of $\hat{r}_B$ in the finite batch and prevents collapse of the correlation objective. We use this corollary only as a non-degeneracy argument for the training loss, not as a claim that the resulting batch correlation is an unbiased or lower-variance estimator of a population correlation. Our experimental practice of stratified sampling across the four speeds is consistent with this corollary. □

### A2.2 Variance decomposition for the stratified $\mathcal{L}_{\text{ratio}}$

**Setup.** A training batch $\mathcal{B}$ of size $B$ partitions into RPM strata $\{\mathcal{B}_r\}_{r\in\mathcal{R}_B}$ with $|\mathcal{B}_r|\geq 2$. Within each stratum, samples share $\omega_p{=}r$ and have i.i.d. $S_C^{(i)} \sim \text{Beta}(\alpha,\beta)$ with $\mu_{S_C}{=}\alpha/(\alpha+\beta)$ and $\sigma^2_{S_C} = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$. For $\text{Beta}(199.5, 10.5)$, $\mu_{S_C}{=}0.95$, $\sigma_{S_C}{=}0.0150$, nominal high-density range approximately $[0.92, 0.98]$.

**Variance of $R_T(S_C)$ within and across strata.** $R_T(S_C)$ is affine in $S_C$ with slope $m \triangleq \sin\theta/(1+\sin\theta) \approx 0.414$ at $\theta{=}45°$, so

$$\text{Var}(R_T)_{\text{within}} \;=\; m^2\sigma^2_{S_C} \;\approx\; 3.86\times10^{-5}. \tag{14}$$

Because $R_T$ depends on $S_C$ but not directly on $\omega_p$, pooling RPM strata does not substantially increase the variance of $R_T$ when the same $S_C$ prior is used in each stratum. The problem is instead that $D^{\text{lat}}$ inherits a much larger speed-driven component: $V_{\text{eff}}$ varies by $\sim$5$\times$ from 21 to 108 RPM, whereas the within-stratum variation of $R_T$ under the real Beta prior is only $m^2\sigma^2_{S_C} \approx 3.86\times10^{-5}$. Decomposing $D^{\text{lat}} = \alpha_v V_{\text{eff}} + \alpha_R R_T + \xi$ with mean-zero $\xi$,

$$\text{Cov}_{\text{pooled}}(D^{\text{lat}}, R_T) = \alpha_R \,\text{Var}(R_T)_{\text{pooled}} + \alpha_v \,\text{Cov}(V_{\text{eff}}, R_T)_{\text{pooled}}. \tag{15}$$

When $S_C$ is sampled independently of RPM, $\text{Cov}(V_{\text{eff}}, R_T)_{\text{pooled}}$ is small in expectation relative to the RPM-driven variance in $V_{\text{eff}}$, while the Pearson denominator for $D^{\text{lat}}$ is dominated by $\alpha_v^2\,\text{Var}(V_{\text{eff}})_{\text{pooled}}$. Thus the pooled correlation can underweight the weak slip-axis variation, motivating the within-RPM stratified loss.

**Mean-normalized within-RPM stratification.** Let $\bar{D^{\text{lat}}}_{\mathcal{B}_r} \triangleq |\mathcal{B}_r|^{-1}\sum_{i\in\mathcal{B}_r} D_i^{\text{lat}}$ and $\bar{R}_{T\,\mathcal{B}_r}$ analogously, and define $\tilde{D}^{(i)} = D_i^{\text{lat}}/\bar{D^{\text{lat}}}_{\mathcal{B}_r}$, $\tilde{R}^{(i)} = R_T(S_C^{(i)})/\bar{R}_{T\,\mathcal{B}_r}$. Within a stratum $V_{\text{eff}}$ is nearly constant because the RPM is fixed and the remaining $S_C$-dependence through $V_C$, $\omega_c$, and $\omega_b$ is small under $\sigma_{S_C}{=}0.015$. So, the dominant axis of within-stratum variation in $D^{\text{lat}}$ is whatever $S_C$-dependence the model produces. Mean-normalization removes the absolute scale, and $\mathcal{L}_{\text{ratio}}$ (Eq. 5) penalizes within-stratum order mismatches between latent dynamics and analytical $R_T$.

- *Why MSE rather than Pearson within the stratum?* For $|\mathcal{B}_r|{=}2$, sample Pearson is $\pm1$ with probability 1 and its gradient is degenerate; MSE between mean-normalized signals stays smooth.
- *Why mean-normalization rather than centering?* Both remove the across-stratum offset; mean-normalization additionally normalizes within-stratum scale, keeping $\mathcal{L}_{\text{ratio}}$ dimensionless and comparable across strata. Centering would re-introduce an RPM-dependent weight per stratum.

**Filtering criterion.** We exclude strata with $|\mathcal{B}_r| < 2$ or $\text{Var}_{\mathcal{B}_r}(R_T) \leq \varepsilon_{S_C}$, since these cannot contribute slip information. The stratified form is therefore a design constraint that prevents RPM-dominated variance from swamping the slip gradient at our batch size, not a separate proof of slip recovery.

### A2.3 Latent–pixel consistency diagnostic

**Setup.** Let $\mathcal{E} : \mathbb{R}^{T\times3\times H\times W} \to \mathbb{R}^{T\times C\times h\times w}$ be the frozen VAE encoder. We define

$$\begin{aligned} D_i^{\text{pix}} &= \frac{1}{(T-1)\cdot 3HW}\sum_{t=1}^{T-1}\left\|\mathbf{v}_i^{(t+1)} - \mathbf{v}_i^{(t)}\right\|_1, \\ D_i^{\text{lat}} &= \frac{1}{(T-1)Chw}\sum_{t=1}^{T-1}\left\|\mathbf{z}_i^{(t+1)} - \mathbf{z}_i^{(t)}\right\|_1, \end{aligned} \tag{16}$$

where $\mathbf{z}_i = \mathcal{E}(\mathbf{v}_i)$. Because the diffusion model is trained in latent space, the physics losses use $D^{\text{lat}}$ rather than optical flow or pixel differences. The required contract is therefore not pixel-level physical fidelity, but preservation of the ordering of clip-level motion magnitudes.

**Diagnostic criterion.** We test this contract empirically by measuring the Pearson correlation between $D^{\text{pix}}$ and $D^{\text{lat}}$ on held-out real clips. A high positive correlation indicates that clips with larger pixel-level temporal change also tend to have larger latent temporal change, making $D^{\text{lat}}$ a suitable

Table 5: Empirical Pearson $r$ between $D^{\mathrm{pix}}$ and $D^{\mathrm{lat}}$ (Sec. 5.3).

| Split / band | $r$ | $p$-value |
|---|---|---|
| Overall ($n$=120, four RPMs pooled) | $\mathbf{0.907}$ | $3.30\times10^{-46}$ |
| RPM 21 (within-group) | 0.778 | n/a |
| RPM 48 (within-group) | 0.827 | n/a |
| RPM 78 (within-group) | 0.939 | n/a |
| RPM 108 (within-group) | 0.920 | n/a |
| Low band (2×2 pooled, circulation signal) | 0.985 | n/a |
| High band (residual, self-rotation signal) | 0.730 | n/a |

scalar proxy for the ordering objective used by $\mathcal{L}_{\mathrm{dyn}}$. This diagnostic does not prove that latent dynamics recover contact mechanics; it only verifies that the latent scalar preserves the dominant ordering of visible motion on the real clips used in this study.

**Empirical instantiation.** We sampled $n$=120 real polishing clips total (30 per RPM) of $T$=14 frames, encoded them through the frozen SVD-xt VAE, and measured $D^{\mathrm{pix}}$ and $D^{\mathrm{lat}}$ per clip. Table 5 reports the resulting correlations. As a reproducibility check, recomputing $V_{\mathrm{eff}}$ with the surface-speed norm in Eq. (2) preserves the ordering of prior diagnostic runs (Pearson $r$=0.997 with the earlier audit target), while providing the dimensionally correct physical scale used in this paper.

The overall correlation $r$=0.907 supports using $D^{\mathrm{lat}}$ as a clip-level motion-ordering proxy for $\mathcal{L}_{\mathrm{dyn}}$. The lower high-band correlation $r$=0.730 also exposes a limitation: latent frame differences preserve circulation like motion more reliably than residual self-rotation. This is why Sec. 5.3 states the result as an empirical scope condition rather than as a guarantee of full contact-mechanics recovery.

## A3 Additional Experimental Results

This supplementary section expands the compact results in Sec. 6. The main paper reports the primary design, synthetic, and LOSO results; here we include the effective-batch-size and Fourier-bandwidth sweeps referenced in Sec. 6.2, along with metric definitions and controlled sweep details.

### A3.1 Metric definitions

**I3D FVD.** We use RGB-I3D features trained on Kinetics-400 [32]. Pre-processing is fixed for all variants: resize to $224 \times 224$, sample 16 frames per clip by bilinear time interpolation from the 30-frame videos, and compute the Fréchet distance between generated and reference feature distributions.

**FID and paired image metrics.** FID [11] is computed with the same RGB-I3D backbone used for FVD so relative comparisons do not mix feature spaces. PSNR, SSIM, and LPIPS [42] are paired against the held-out clip associated with the same first frame.

**Diagnostic flow correlation.** A diagnostic Pearson correlation between RPM and mean per-clip Farneback flow magnitude [7] is computed for sanity-checking, but the headline speed and slip metrics are the gated rank accuracies recovered through the tracking pipeline (Sec. 6.1).

### A3.2 Alternative motion signals for $\mathcal{L}_{\mathrm{dyn}}$/$\mathcal{L}_{\mathrm{ratio}}$

We also checked optical-flow magnitude as a candidate dynamics signal. Table 6 summarizes the diagnostic correlations and controlled slip-ranking baselines. On the same 120 real clips used for the latent–pixel diagnostic, Farneback mean flow magnitude correlates strongly with pixel L1 ($r$=0.933), latent L1 ($r$=0.833), and the corrected surface-speed $V_{\mathrm{eff}}$ ($r$=0.905). Thus flow is a good aggregate-motion diagnostic. It does not by itself solve slip supervision, however: a scalar flow magnitude measures how much the image moves, not whether the motion follows the slip-sensitive circulation ratio $R_T(S_C)$.

The flow-magnitude warp is intentionally slip-flat: it matches the mean flow magnitude at each RPM and therefore answers the reviewer concern that the model might only be learning speed. Its rank accuracy is at chance. The analytical warp uses the Lee kinematics directly and activates the slip

Table 6: Alternative motion-signal diagnostics for $\mathcal{L}_{\mathrm{dyn}}$ and $\mathcal{L}_{\mathrm{ratio}}$.

| Signal / baseline | Diagnostic | Value |
|---|---|---|
| Farneback flow vs. $D^{\mathrm{pix}}$ | Pearson $r$ on real clips | $0.933$ |
| Farneback flow vs. $D^{\mathrm{lat}}$ | Pearson $r$ on real clips | $0.833$ |
| Farneback flow vs. corrected $V_{\mathrm{eff}}$ | Pearson $r$ on real clips | $0.905$ |
| Flow-magnitude warp | Controlled slip rank accuracy | $0.484$ |
| Analytical kinematic warp | Controlled slip rank accuracy | $0.804$ |

axis, but it is a hand-coded image warp and has weaker paired fidelity than the five-seed diffusion model row in Table 4 (PSNR $12.37$ and SSIM $0.591$ versus $16.69$ and $0.689$). Learned flow or video embeddings such as RAFT [30] and VideoMAE [31] are plausible future signals if they retain direction and track geometry, but a scalar learned flow magnitude would likely inherit the same limitation as Farneback magnitude. Complete $\mathcal{L}_{\mathrm{dyn}}/\mathcal{L}_{\mathrm{ratio}}$ variants with these proxies would require decoding latents or adding a frozen video encoder inside the training loop and then calibrating a vector feature to scalar physics targets. The latent $\ell_1$ signal keeps the loss cheap, differentiable almost everywhere, and aligned with the SVD latent space while the analytical $V_{\mathrm{eff}}, R_T$ targets provide the missing physical direction.

### A3.3 Tracking and kinematic inversion pipeline

The evaluation pipeline translates generated video clips into estimated kinematic quantities $(\widehat{\omega_c},\ \widehat{S_C})$ using only pixel observations, with no access to latent codes or training signals. This independence is essential: a pipeline that reused training-time quantities would not constitute an external physical test of controllability.

**Step 1: Detection and tracking.** A YOLO11 detector fine-tuned on polishing footage localises shell centroids in each frame. Detections are linked across frames by a Mahalanobis-IOU hybrid tracker following [44], which uses detection confidence to weight the association cost. Each track $k$ is a sequence of frame-indexed centroids and uncertainty estimates $\{(t_n,\ c_{x,n},\ \sigma_{x,n})\}_{n=1}^{N_k}$.

**Step 2: Per-track angular velocity fitting.** Because the shell orbits in a plane, its $x$-centroid follows a cosine projection of the orbit angle. Inverting that projection and fitting the unwrapped angle linearly in time gives an unbiased angular speed estimate without requiring the full 2-D orbit to be visible.

For track $k$, project the centroid onto the orbit angle:

$$u_n = \frac{c_{x,n} - x_{\mathrm{ctr}}}{r_{\mathrm{orb}}}, \qquad \theta_n = \cos^{-1}(u_n), \tag{17}$$

where $x_{\mathrm{ctr}}$ and $r_{\mathrm{orb}}$ are the orbit centre and radius estimated from the suite geometry. The raw $\arccos$ introduces a reflection ambiguity at the orbit poles; this is resolved by a sign-of-$\Delta x$ unwrapping step that gives a monotonically increasing angle series $\tilde{\theta}_n$. A weighted ordinary-least-squares fit then yields the signed angular speed:

$$\hat{\omega}^{(k)} = \arg\min_{a,b} \sum_n \big(\tilde{\theta}_n - a t_n - b\big)^2. \tag{18}$$

The fit quality is assessed by $R^2 = 1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$. Four per-track gates remove unreliable estimates before aggregation: (i) minimum track length $N_k \geq N_{\min}$; (ii) near-singularity fraction $|\{n : |u_n| > u^*\}|/N_k \leq f^*$ (tracks that spend most of their time near the arccos poles have large projection noise); (iii) minimum angular range $\max_n \tilde{\theta}_n - \min_n \tilde{\theta}_n \geq \Delta\theta_{\min}$; (iv) minimum fit quality $R^2 \geq R^2_{\min}$.

**Step 3: Cross-track aggregation.** Even after per-track gating, a small number of spurious tracks can bias a simple mean. We therefore apply a two-stage robust aggregation [27].

First, a median absolute deviation (MAD) outlier screen removes tracks whose slope deviates from the per-clip median by more than $c$ scaled MADs:

$$\mathrm{MAD} = \mathrm{median}_k\big(|\hat{\omega}^{(k)} - \mathrm{median}_k\, \hat{\omega}^{(k)}|\big)\ ,\ \text{keep track } k \iff \big|\hat{\omega}^{(k)} - \mathrm{median}_k\, \hat{\omega}^{(k)}\big| \leq c \cdot 1.4826 \cdot \mathrm{MAD} \tag{19}$$

Second, a weighted-median aggregation over the surviving set $\mathcal{K}$ uses weights $w_k = N_k \cdot \max(R_k^2, 0)$ to up-weight long, well-fit tracks:

$$\widehat{\omega_c} = \text{sgn}(\text{majority}) \cdot |\hat{\omega}_{(\pi)}|, \quad \pi = \min\Big\{k : \sum_{j \le k} w_{(j)} \ge \tfrac{1}{2} \textstyle\sum_j w_j\Big\}, \tag{20}$$

where $(j)$ indexes tracks sorted by $|\hat{\omega}|$ and the majority sign is determined by sign-consistency voting across $\mathcal{K}$.

**Step 4: Physical validity gate.** The kinematic inversion in Step 5 is only interpretable when the recovered $\widehat{\omega_c}$ lies within the physically plausible range implied by the Lee model [19] for the suite's operating conditions. Clips whose $\widehat{\omega_c}$ falls outside this window are excluded from rank accuracy; reporting only on plausible clips prevents out-of-regime tracker noise from polluting the controllability scores.

From the Lee kinematics, the circulation ratio is

$$R_T(S_C, \theta) = \frac{1 + S_C \sin\theta}{1 + \sin\theta}, \qquad \omega_c = \omega_p \cdot R_T, \tag{21}$$

giving a plausible window

$$0.5\, \omega_{p,\min} R_{T,\min} \;\le\; |\widehat{\omega_c}| \;\le\; \kappa\, \omega_{p,\max} R_{T,\max}, \tag{22}$$

where $R_{T,\min\,/\,\max}$ are evaluated at the suite's slip and RPM extremes. A clip passes the gate when Eq. (22) holds. Averaged over five seeds, clip-level gate yield ranges from 93.4% to 97.5% across the nine real-regime variants in Table 1 and from 66.6% to 68.8% across the four synthetic-regime variants in Table 2, which now report yield per row. This is distinct from the within-clip track-retention rate above (the fraction of individual tracks, not clips, kept after the length, singularity, $R^2$, and cross-track MAD checks), which is lower, around 28%–32% in the real regime; an earlier draft reported this track-retention figure as gate yield, which was an error.

**Step 5: Kinematic inversion.** Given the aggregated $\widehat{\omega_c}$ and the known operating speed $\omega_p$, the Lee equations are inverted to recover the circulation ratio and then the contact slip:

$$\widehat{R_T} = \frac{\widehat{\omega_c}}{\omega_p}, \qquad \widehat{S_C} = \frac{\widehat{R_T}\,(1 + \sin\theta) - 1}{\sin\theta}. \tag{23}$$

A single scalar calibration coefficient $c_{\text{suite}}$, estimated on a validation split without test labels, corrects small geometric bias in $\widehat{\omega_c}$ before inversion; no slip labels enter this correction.

**Step 6: Rank accuracy.** A surrogate is controllable along a given axis if it can *rank* operating conditions correctly, even if absolute calibration is imperfect. Rank accuracy is therefore the natural metric: it equals 1 when generated clips are perfectly ordered by the conditioning variable, 0.5 at chance, and is insensitive to monotone rescaling of $\hat{y}$.

Formally, for a group $g$ of clips sharing the confounding variables (i.e., same scene and slip for the speed axis; same scene and RPM for the slip axis), we compute the pairwise concordance [18]:

$$\text{RankAcc}_g = \frac{\big|\{(i,j) : i < j,\ (\text{gt}_i - \text{gt}_j)(\hat{y}_i - \hat{y}_j) > 0\}\big|}{\big|\{(i,j) : i < j,\ \text{gt}_i \neq \text{gt}_j\}\big|}. \tag{24}$$

The two evaluation axes use different ground-truth and estimate pairs:

| Axis | Group by | gt | $\hat{y}$ |
|---|---|---|---|
| Speed | (scene, $S_{C,\text{gt}}$) | $\omega_p$ (RPM) | $\widehat{\omega_c}$ |
| Slip | (scene, RPM) | $R_{T,\text{gt}}$ | $\widehat{R_T}$ |

The reported value is the bootstrap mean of $\{\text{RankAcc}_g\}$ over groups, with 95% percentile-bootstrap confidence intervals (1000 resamples).

### A3.4 Controlled slip sweep and physical-validity gate

For controllability, each variant generates 64 clips per seed: four scenes, four RPM values, and four slip values ($S_C \in \{0.92, 0.94, 0.96, 0.98\}$ on $R$, $\{0.20, 0.40, 0.60, 0.80\}$ on $S$). The full tracking and

inversion procedure of Sec. A3.3 is run identically on every generated clip. A track is physically valid (plausibility gate, Eq. (22)) when $\widehat{R_T} = \widehat{\omega_c}/\omega_p \in [0.8, 1.2]$. The gate is intentionally conservative: it reduces sample size but prevents tracks outside the Lee-model regime from driving the controllability scores.

### A3.5 Effective-batch-size and Fourier-bandwidth sweeps

Table 7 reports the two recipe sweeps referenced in Sec. 6.2.

Table 7: Effective-batch-size and Fourier-bandwidth sweeps on $R$, mean over five seeds, plausibility gate. **Bold** marks the best entry within each group. Default configuration is marked $^\dagger$.

| | | Perceptual fidelity | | | | | Speed axis | | Slip axis | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Group** | **Variant** | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ | FVD↓ | rank↑ | $\rho_s$ ↑ | rank [95% CI]↑ | $\rho_s$ ↑ |
| $B_{\text{eff}}$ | 2 | 17.52 | 0.714 | 0.166 | 162.0 | 192.7 | 0.975 | 0.907 | 0.492 [.36,.62] | -0.030 |
| | 4 | **17.68** | **0.715** | **0.166** | 161.0 | **188.9** | 0.977 | 0.915 | **0.513** [.38,.64] | **+0.047** |
| | 8$^\dagger$ | 17.10 | 0.700 | 0.173 | 158.0 | 192.6 | 0.967 | 0.908 | 0.492 [.36,.63] | +0.004 |
| | 16 | 16.38 | 0.684 | 0.181 | **156.9** | 192.4 | 0.975 | **0.916** | 0.479 [.35,.61] | -0.010 |
| $K$ | 4 | 16.54 | 0.685 | 0.180 | 160.2 | 190.3 | 0.969 | 0.902 | 0.498 [.38,.62] | **+0.032** |
| | 8 | 15.95 | 0.671 | 0.193 | 159.1 | **186.2** | 0.971 | 0.900 | 0.503 [.38,.63] | +0.026 |
| | 16$^\dagger$ | **16.70** | **0.690** | **0.174** | 161.9 | 194.4 | 0.971 | **0.918** | **0.517** [.38,.65] | +0.021 |
| | 32 | 10.88 | 0.439 | 0.356 | 219.0 | 264.5 | 0.770 | 0.624 | 0.441 [.24,.65] | -0.098 |

### A3.6 Qualitative generated frames across speeds

Sec. 6 reports frame-level and video-level metrics but shows only one representative frame per condition (Fig. 5). To let a reader judge temporal coherence directly, Figs. 6 and 7 show every frame of a generated clip at each of the four RPMs, for the full model (both physics losses, cross attention, pretrain plus physics fine tune) at its final training epoch. These clips come from the model's saved qualitative sample and validation outputs at the final epoch, at a thin top-down crop resolution ($496 \times 48$ pixels for $R$, $480 \times 48$ for $S$) rather than full evaluation resolution. Frames keep their native aspect ratio; each speed block lays its 30 frames out in a $10 \times 3$ grid rather than one long row, so individual frames stay large and readable without any stretching. The synthetic clips use $S_C$=0.50, a value from training-time monitoring rather than the $\{0.20, 0.40, 0.60, 0.80\}$ evaluation grid in Sec. 6.3. Across both figures, the capsule contact point moves smoothly from frame to frame with no visible flicker or identity swap, and it advances further per frame at higher RPM, consistent with the speed axis ranking in Table 1.

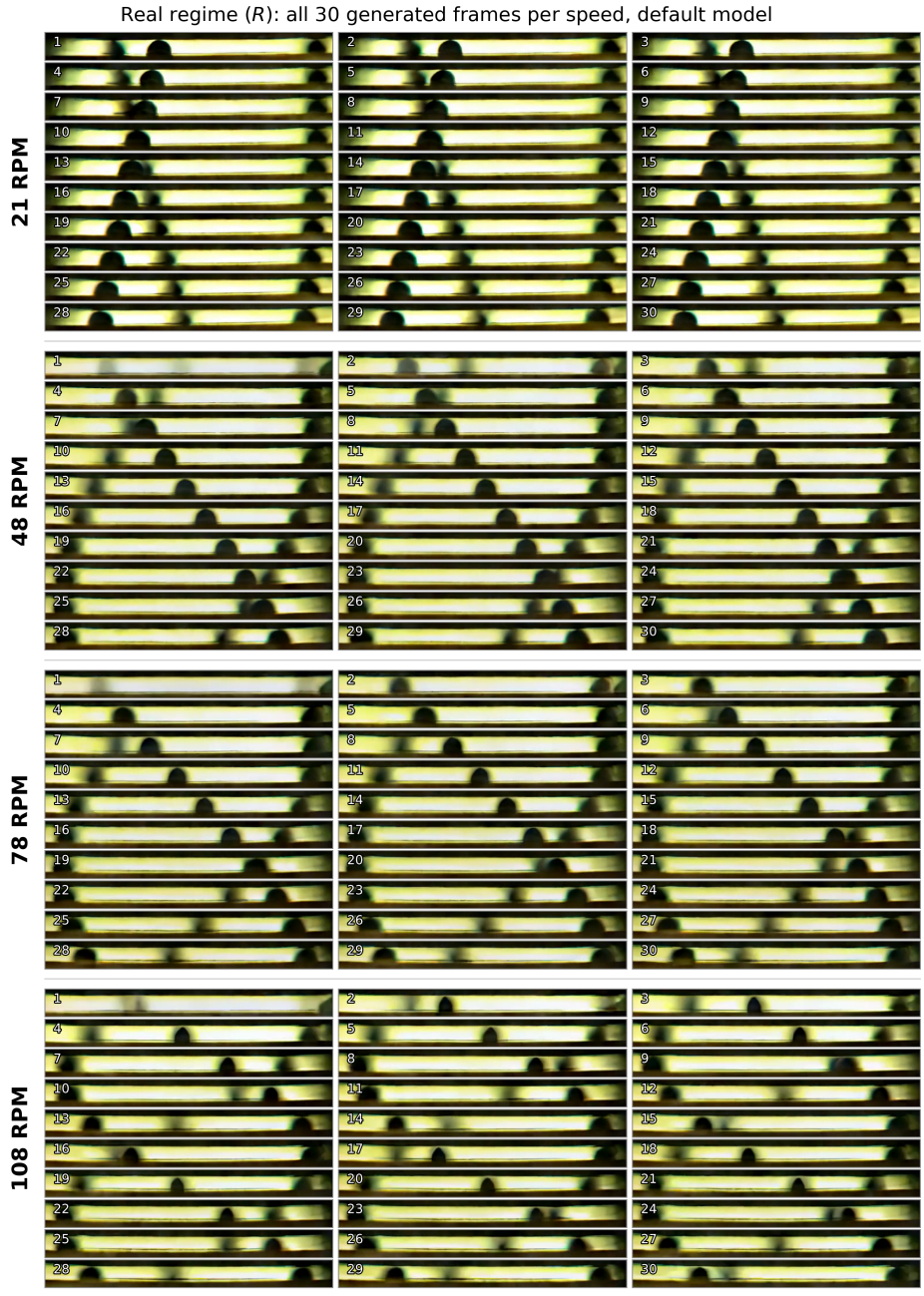



Figure 6: All 30 frames of the full model's generated clip at each real-regime ($R$) speed, read left to right then top to bottom within each speed block.

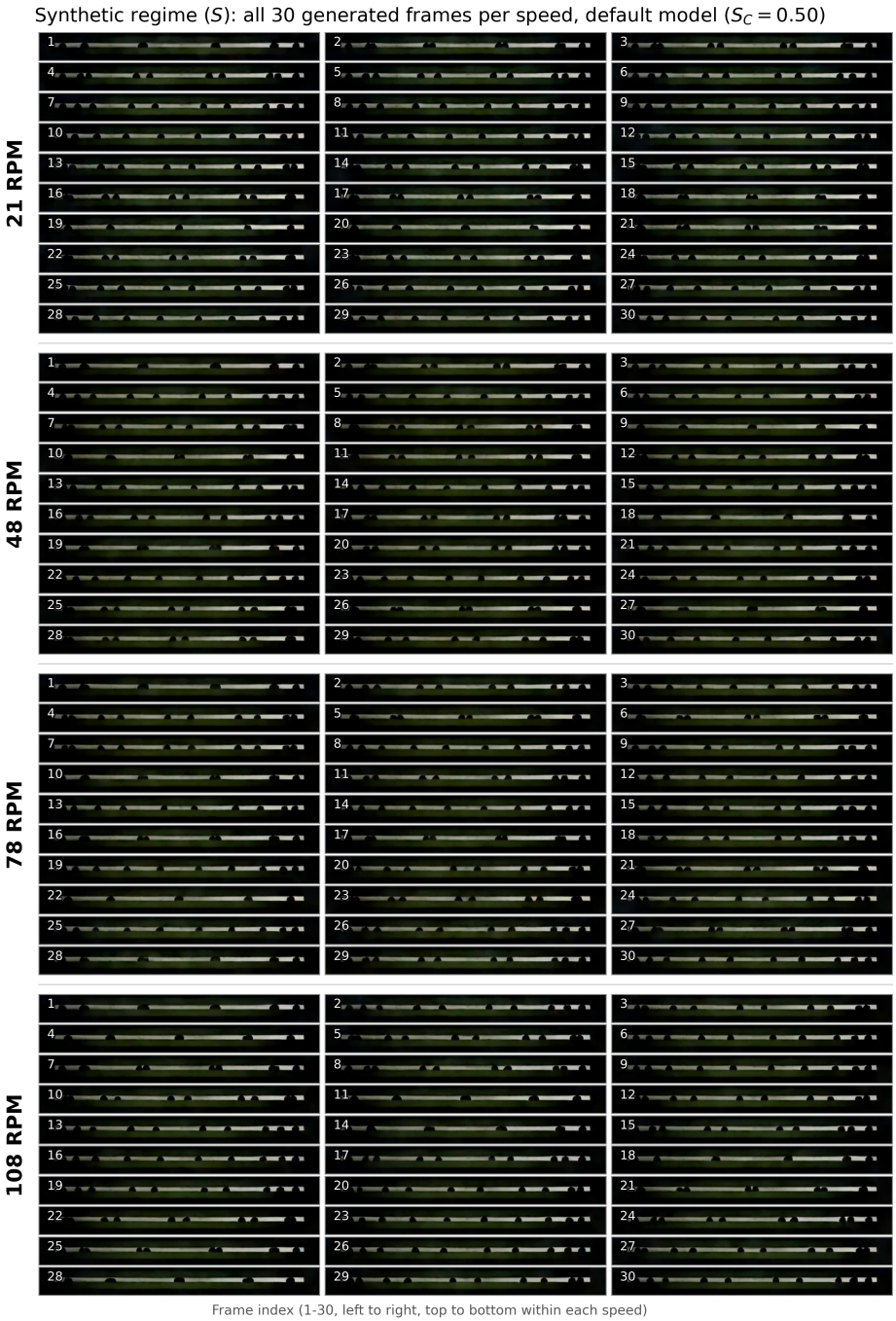



Figure 7: All 30 frames of the full model's generated clip at each synthetic-regime ($S$) speed, at $S_C$=0.50, read left to right then top to bottom within each speed block.

## A4 Training Details

### A4.1 Hyperparameters

Table 8 summarizes the training and inference hyperparameters used for the main experiments.

Table 8: Training and inference hyperparameters.

| Parameter | Value |
|---|---|
| **Optimization** | |
| Optimizer | AdamW |
| Weight decay | 0.01 |
| LoRA learning rate | $1 \times 10^{-4}$ |
| Physics-encoder learning rate | $5 \times 10^{-4}$ |
| LR schedule | Cosine + 500-step warmup |
| Per-GPU batch size | 2 (default real-design runs) |
| Gradient accumulation | 4 steps (default real-design runs) |
| Effective batch size | 8 default; audited per suite below |
| LoRA rank $r$ | 32 |
| Epochs | 50 (per ablation script) |
| GPUs | single RTX 4090 per run; multiple runs in parallel across GPUs |
| **Conditioning** | |
| Number of physics tokens | 3 (RPM, $S_C$, interaction) |
| Fourier bands $K$ for each scalar | 16 |
| CFG drop probability $p_{\text{cfg}}$ | 0.1 |
| Token dim $d$ | 1024 (matches SVD-xt CLIP embedding) |
| **Two-phase schedule** | |
| Phase-1 epochs (perceptual only) | 10 |
| Ramp window (linear $w(e)$) | 5 epochs |
| **Loss weights (full config)** | |
| $\lambda_r$ (reconstruction) | 1.0 |
| $\lambda_s$ (SSIM) | 0.5 |
| $\lambda_d$ (dynamics, $\mathcal{L}_{\text{dyn}}$) | 0.05 |
| $\lambda_t$ (temporal, $\mathcal{L}_{\text{temp}}$) | 0.08 |
| $\lambda_\rho$ (ratio, $\mathcal{L}_{\text{ratio}}$) | 0.05 (full); 0.0 elsewhere |
| $\lambda_o$ (orthogonality) | 0.01 |
| $\lambda_c$ (contrastive) | 0.01 |
| **Inference** | |
| Sampler | DDIM [28] |
| Steps | 25 |
| CFG scale $s$ | $\{3.0, 5.0\}$ (reported per-experiment) |

**Effective-batch-size audit.** The default $B_{\text{eff}}$=8 in Table 8 refers to the real-data design-study configuration used for the main ablations and the $B_{\text{eff}}$ sweep in Table 7. Table 9 reports the effective batch sizes used by the newer long-run suites.

Table 9: Effective batch sizes used across run families.

| Run family | Per-device | Accum. | $\mathbf{B}_{\text{eff}}$ |
|---|---|---|---|
| Real design study default / 50 epoch | 2 | 4 | 8 |
| Real $B_{\text{eff}}$ sweep (A5) | varies | varies | $\{2, 4, 8, 16\}$ |
| Real full-scale 100-epoch suite | 4 | 3 | 12 |
| Real LOSO 100-epoch suite | 4 | 3 | 12 |
| Synthetic ($S$) loss sweep / LOSO 100-epoch suite | 4 | 4 | 16 |

### A4.2 Auxiliary loss definitions

**Orthogonality regularizer.** $\mathcal{L}_{\text{orth}}$ pulls the two independent physics tokens $\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C}$ toward orthogonality so that the RPM and slip axes are encoded in separable subspaces:

$$\mathcal{L}_{\text{orth}} \;=\; \big|\cos(\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C})\big|. \tag{25}$$

**Soft contrastive regularizer.** $\mathcal{L}_{\text{contr}}$ pulls RPM tokens of clips with similar $V_{\text{eff}}$ closer in embedding space:

$$\mathcal{L}_{\text{contr}} \;=\; \frac{1}{B(B-1)} \sum_{i \neq j} w_{ij} \, \|\mathbf{T}_{\text{rpm}}^{(i)} - \mathbf{T}_{\text{rpm}}^{(j)}\|_2^2, \qquad w_{ij} \;=\; \exp\!\big(-|V_{\text{eff}}^{(i)} - V_{\text{eff}}^{(j)}|\big). \tag{26}$$

### A4.3 Compute budget

Table 10 summarizes the compute resources used for training and inference.

Table 10: Compute budget and trainable parameter count.

| Resource | Value |
|---|---|
| **Training** | |
| Hardware | Single RTX 4090 (24 GB) |
| GPU-hours (50-epoch) | $\approx$ 6–8 h |
| GPU-hours (100-epoch) | $\approx$ 12–16 h |
| Total (all ablations) | $\approx$ 500 GPU-hours |
| Peak GPU memory | $\approx$ 20 GB |
| **Inference** | |
| Resolution | $496 \times 48$ (raw $2040 \times 204$, resize 0.25, aligned to 16×) |
| Frames | 30 |
| Latency per clip (DDIM 25-step) | $\approx$ 35–45 s |
| Throughput | $\approx$ 80–100 clips/GPU-h |
| **Trainable parameters (LoRA + physics encoder)** | |
| SVD-xt backbone (frozen) | 1.5 B |
| LoRA (rank 32, all attn. layers) | $\approx$ 22 M |
| Physics encoder (MLPs + interaction) | $\approx$ 2 M |
| Total trainable | $\approx$ 24 M (1.6% of backbone) |

### A4.4 Latent–pixel correlation protocol

For the latent–pixel diagnostic of Sec. 5.3, we sample 120 real polishing clips total (30 at each of the four training RPMs) with $T$=14 frames, compute $D^{\text{pix}}$ and $D^{\text{lat}}$ per clip, and report Pearson $r$. Preprocessing mirrors the training path: full frame, resize factor 0.25, pixel values in $[0, 1]$, and direct VAE encoding without a $2x - 1$ pixel shift.

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